Predicting Diabetes Risk Using BRFSS Health Indicators with Explainable Artificial Intelligence

Final Research Paper - Phase 3

Team 13
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Abstract

Diabetes is a major chronic disease that can lead to cardiovascular disease, kidney failure, neuropathy, vision loss, and premature mortality. Early risk prediction can support screening and preventive care, but machine learning models in healthcare must be interpretable because unexplained predictions are difficult to trust in clinical decision support. This project develops an explainable artificial intelligence framework for predicting diabetes status using the 2015 Behavioral Risk Factor Surveillance System (BRFSS) dataset. The final task is a three-class classification problem: no diabetes, prediabetes, and diabetes. The raw BRFSS file was cleaned by removing invalid survey responses, recoding the diabetes target, transforming coded variables, engineering clinically meaningful indicators, and selecting twenty predictors. Six paper-inspired models were implemented: multilayer perceptron (Xie et al.), XGBoost (Ullah et al.), histogram gradient boosting (Chowdhury et al.), Extra Trees (Islam et al.), a TIPNet-inspired deep tabular model (Zafar et al.), and CatBoost (Maimaitijiang et al.). Models were evaluated using accuracy, balanced accuracy, macro-F1, weighted-F1, diabetes recall, ROC-AUC, and confusion matrices. Each model was explained using permutation feature importance, partial dependence plots, individual conditional expectation plots, SHAP, LIME, and global surrogate trees. Results show that the best model depends on the objective: the MLP achieved the highest overall accuracy, XGBoost and histogram gradient boosting achieved stronger diabetes recall, and CatBoost achieved the strongest weighted-F1 among Youssef's models. Across XAI methods, general health, high blood pressure, BMI-related variables, age, high cholesterol, and checkup frequency repeatedly appeared as influential predictors. The study demonstrates that diabetes prediction should be evaluated as both a performance problem and an explanation problem, especially under class imbalance.

1 Introduction

Diabetes mellitus is a chronic disease characterized by abnormal blood glucose regulation. It is associated with serious long-term complications, including cardiovascular disease, kidney disease, nerve damage, vision problems, and increased mortality. Because type 2 diabetes develops gradually and is linked to many behavioral, demographic, and health-condition factors, early screening can help identify individuals who may benefit from follow-up testing or preventive intervention. However, a prediction model in healthcare is not useful only because it produces a label. It must also provide understandable reasoning so that model behavior can be inspected, challenged, and communicated responsibly.

This project investigates diabetes-status prediction using the 2015 Behavioral Risk Factor Surveillance System (BRFSS), a large public survey dataset collected by the Centers for Disease Control and Prevention (CDC). BRFSS contains self-reported health indicators such as BMI, general health, high blood pressure, high cholesterol, physical activity, healthcare access, income, age group, and related lifestyle variables. Unlike laboratory datasets that include glucose or insulin values, BRFSS uses survey indicators. This makes the task more challenging, but also useful for population-level screening because survey data are easier to collect at scale than laboratory measurements.

The implementation includes a shared preprocessing and EDA notebook, feature engineering, feature selection, six paper-inspired model notebooks, model evaluation, and explainability outputs.

The main objective is to compare machine learning and deep learning models not only by predictive performance but also by their interpretability. The project therefore evaluates the following research questions:

1. Which implemented model provides the strongest overall classification performance on the three-class BRFSS diabetes prediction task?
2. Which model provides the most useful diabetes-class recall for screening-oriented use?
3. Which health indicators are consistently highlighted by different XAI methods?
4. How do global, local, and surrogate-based explanation techniques complement each other?
5. How does severe class imbalance affect the interpretation of accuracy and clinical usefulness?

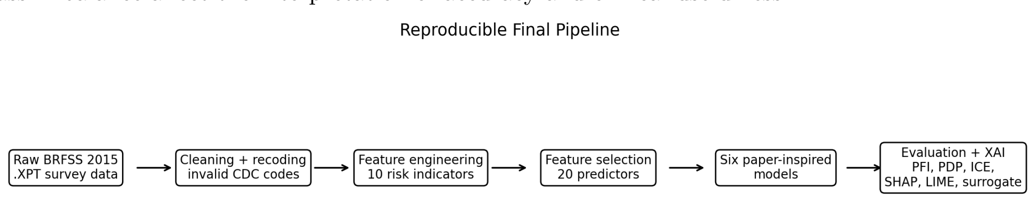


Figure 1: Final project pipeline. The workflow begins with the raw BRFSS survey file, applies cleaning and feature engineering, selects final predictors, trains six paper-inspired models, and explains predictions using multiple XAI tools.

2 Related Work

The literature on diabetes prediction can be organized into five major themes: survey-based risk prediction, imbalance-aware learning, explainable tabular machine learning, deep learning for diabetes prediction, and deployable decision-support systems. This section expands the Phase 1 literature review and connects previous work directly to the final implementation decisions.

2.1 Survey-Based Diabetes Risk Prediction

Xie et al. developed risk prediction models for type 2 diabetes using BRFSS survey data and compared support vector machine, decision tree, logistic regression, random forest, neural network, and Gaussian Naive Bayes classifiers [?]. Their neural network had the highest accuracy and AUC, while the decision tree had the highest sensitivity. This difference is important because a model that is best for overall accuracy may not be best for screening. Our Xie-inspired track therefore implements an MLP but evaluates it using both accuracy and diabetes recall.

BRFSS-based modeling is valuable because it allows large-scale screening using behavioral and demographic variables. However, survey data include self-reporting bias, missingness, coded response categories, and class imbalance. Xie et al. also identified checkup frequency and sleeping time as potential additional risk factors. In our dataset, checkup-related variables and general health remain important in feature importance and XAI outputs.

2.2 Imbalance-Aware Diabetes Prediction

Ullah et al. focused on detecting high-risk factors and early diabetes diagnosis using a processed BRFSS dataset [?]. They applied SMOTE-ENN to handle class imbalance and reported strong performance for KNN and other machine learning methods. Their main contribution is the emphasis on balancing minority diabetes cases before model training. In our experiments, however, aggressive SMOTE-ENN was unstable under the natural imbalanced test distribution and caused very low accuracy in some notebooks. Therefore, our final Ullah-inspired model uses XGBoost with a more stable imbalance-aware setup rather than relying on an aggressive resampling pipeline.

Chowdhury et al. studied machine learning algorithms and data augmentation methods for class-imbalanced BRFSS diabetes diagnosis [?]. Their paper compared oversampling, undersampling, and hybrid sampling techniques such as SMOTE-N, Edited Nearest Neighbor, SMOTE-Tomek, and SMOTE-ENN. Their work is important because it stresses that sampling must be applied only on the training data to avoid leakage. Our Chowdhury-inspired implementation preserves the imbalanced-data focus but uses histogram gradient boosting instead of destructive resampling because the final test distribution is naturally imbalanced.

Netayawijit et al. proposed a diabetes prediction framework combining SMOTE and SHAP for clinical decision support [?]. Their work shows that imbalance handling and explainability should be integrated rather than treated as separate parts. This motivates our decision to report both performance metrics and explanation outputs for every model.

2.3 Explainable Machine Learning in Diabetes Prediction

Pang compared several explainable machine learning models for diabetes prediction and emphasized Shapley-value interpretation [?]. Although this final paper excludes Ehab's implementation, Pang remains important in the literature because it demonstrates how model comparison can be connected to feature attribution. Ahmed et al. compared LIME and SHAP for explainable diabetes prediction [?]. Their work supports the idea that no single explanation method is sufficient: SHAP is useful for stable global and local attribution, while LIME is useful for explaining individual predictions in a simpler local format.

Islam et al. proposed an explainable machine learning framework for diabetes prediction using random oversampling, quantile transformation, hyperparameter tuning, SHAP, partial dependency, and LIME [?]. They reported Extra Trees as a strong model on multiclass BRFSS data. This paper directly informs Youssef's Extra Trees implementation. Our work extends the explanation set by also using permutation feature importance, ICE plots, and global surrogate trees.

2.4 Deep Learning and Deployment-Oriented Diabetes Systems

Zafar et al. proposed TIPNet, a deep learning model for diabetes prediction that combines temporal and static feature-learning ideas and applies SHAP and LIME for interpretation [?]. Their paper also emphasizes adaptive synthetic oversampling and k-fold validation. In our implementation, the BRFSS feature table does not contain true time-series signals, so the final implementation is a TIPNet-inspired tabular deep architecture rather than an exact reproduction. This is a practical adaptation of the paper idea to the available data format.

Maimaitijiang et al. proposed an online explainable AI framework for diabetes risk prediction using CatBoost, SHAP, SMOTE, and a personalized chatbot assistant [?]. Their work is especially relevant because it moves from offline prediction toward user-facing decision support. Our CatBoost track follows their tabular-learning and explanation direction, but the chatbot component is outside the current scope.

2.5 Explainability and Model Interpretation in the Reviewed Diabetes Studies

The explainability methods used in this project are motivated by the diabetes-prediction studies reviewed in Phase 1 rather than by separate general-purpose XAI papers. Several of the reviewed studies show that prediction performance alone is not sufficient in healthcare settings, because clinicians and patients also need to understand which health indicators drive the model decision. For example, Islam et al. [?] integrated SHAP, partial dependence, and LIME to explain diabetes predictions after model training and hyperparameter tuning. Zafar et al. [?] used SHAP and LIME to interpret the proposed deep learning model and provide insight into the features influencing diabetes classification. Pang [?] focused on Shapley-value-based interpretation to compare how different models reason about diabetes risk. Maimaitijiang et al. [?] used SHAP in an explainable online diabetes-risk prediction framework, while Netayawijit et al. [?] combined imbalance handling with interpretable machine learning for diabetes decision support.

In contrast, earlier BRFSS-based studies such as Xie et al. [?], Ullah et al. [?], and Chowdhury et al. [?] focused more heavily on predictive performance, risk-factor identification, and class-imbalance handling than on modern post-hoc XAI tools. This difference directly shaped our project design. We implemented the strongest or most representative model direction from each selected paper, but we applied a unified explainability framework to every model, even when the original paper did not include all of these explanation techniques. This allows the final comparison to evaluate both predictive behavior and interpretability under the same dataset, preprocessing pipeline, and evaluation setting.

Table 1: Related-work comparison based only on the papers reviewed in Phase 1.

Study	Dataset	Models / Methods	Explainability	Relevance to this project
Xie et al. [?]	BRFSS 2014	SVM, Decision Tree, Logistic Regression, Random Forest, Neural Network, Gaussian Naive Bayes	Regression-based risk-factor analysis; no modern post-hoc XAI	Provides the core BRFSS baseline. The neural network achieved the strongest accuracy/AUC direction, while the decision tree was useful for sensitivity-oriented screening.
Ullah et al. [?]	Processed BRFSS dataset	KNN, Random Forest, XGBoost, Bagging, AdaBoost with SMOTE-ENN balancing	Limited formal XAI	Highlights the importance of imbalance handling in BRFSS diabetes prediction. Our implementation uses this idea but avoids overly destructive balancing when it harms test-distribution accuracy.
Chowdhury et al. [?]	Imbalanced BRFSS dataset	ML classifiers with SMOTE-N, ENN, SMOTE-Tomek, and SMOTE-ENN	Mainly focused on imbalance analysis rather than XAI	Shows that sampling methods must be applied carefully and only on training data to avoid leakage. This supports our decision to compare balancing strategies and report their trade-offs.
Islam et al. [?]	BRFSS multiclass dataset and Diabetes 2019 binary dataset	Extra Trees, Random Forest, Decision Tree, KNN, quantile transformation, random oversampling, GridSearchCV	SHAP, Partial Dependence, LIME	Directly motivates the Extra Trees track and the use of multiple explanation tools. It is one of the closest studies to our final multiclass BRFSS setup.

Study	Dataset	Models / Methods	Explainability	Relevance to this project
Zafar et al. [?]	Diabetes health-indicators dataset	TIPNet, LSTM, Inception-Net, MLP, ADASYN, cross-validation	SHAP and LIME	Motivates the deep-learning track. Our implementation uses a TIPNet-inspired tabular architecture because our final data table is cross-sectional rather than truly temporal.
Maimaitijiang et al. [?]	Diabetes risk dataset with 520 patient records	CatBoost, SMOTE, online risk-prediction interface, personalized assistant	SHAP-based patient-level explanations	Motivates the CatBoost track and demonstrates how explainability can support patient-facing diabetes-risk interpretation. The chatbot component is outside our current scope.
Pang [?]	CDC-derived diabetes data	Logistic Regression, Decision Tree, Random Forest, SVM, KNN, GBM, MARS	Shapley-value explanations	Shows how comparative model evaluation can be combined with feature-attribution analysis. This supports our unified model-comparison and XAI-comparison framework.
Ahmed et al. [?]	Survey-based diabetes prediction dataset	Logistic-regression-based prediction and interpretable modeling	LIME and SHAP	Supports the use of both local and global explanations, especially for simpler models where interpretability and performance can be compared directly.
Netayawijit et al. [?]	Diabetes decision-support dataset	Interpretable machine learning with imbalance handling	SHAP	Supports the combination of class-balancing techniques with explanation methods for healthcare decision support.

Overall, the reviewed papers show three main trends. First, BRFSS-based diabetes prediction is a valid and widely used direction for public-health machine learning. Second, class imbalance is a repeated challenge, especially when the diabetic or prediabetic classes are much smaller than the no-diabetes class. Third, recent work increasingly combines prediction with explainability, especially through SHAP, LIME, and partial-dependence-style analysis. Our project builds on these trends by implementing multiple paper-based models on the same BRFSS 2015 pipeline and applying the same explanation framework across models to enable a fair comparison.

3 Dataset, Preprocessing, and Exploratory Data Analysis

3.1 Dataset Source and Size

The project uses the 2015 BRFSS dataset published by the CDC [?]. The original raw dataset contains 441,456 survey responses and 330 variables stored in SAS Transport format. From the raw file, the preprocessing notebook selected diabetes-related variables covering demographics, lifestyle, healthcare access, physical and mental health, cardiovascular risk, and socioeconomic indicators.

The final usable modeling dataset contained 250,069 records after filtering invalid survey responses, recoding variables, removing missing values, and engineering features. The final train/test split contained 200,055 training records and 50,014 test records.

3.2 Target Definition

The original BRFSS target variable is `DIABETE3`. It was recoded into a three-class target:

- **0 - No diabetes**: respondents who did not report diabetes.
- **1 - Prediabetes/intermediate**: respondents who reported prediabetes or borderline diabetes.
- **2 - Diabetes**: respondents who reported diabetes.

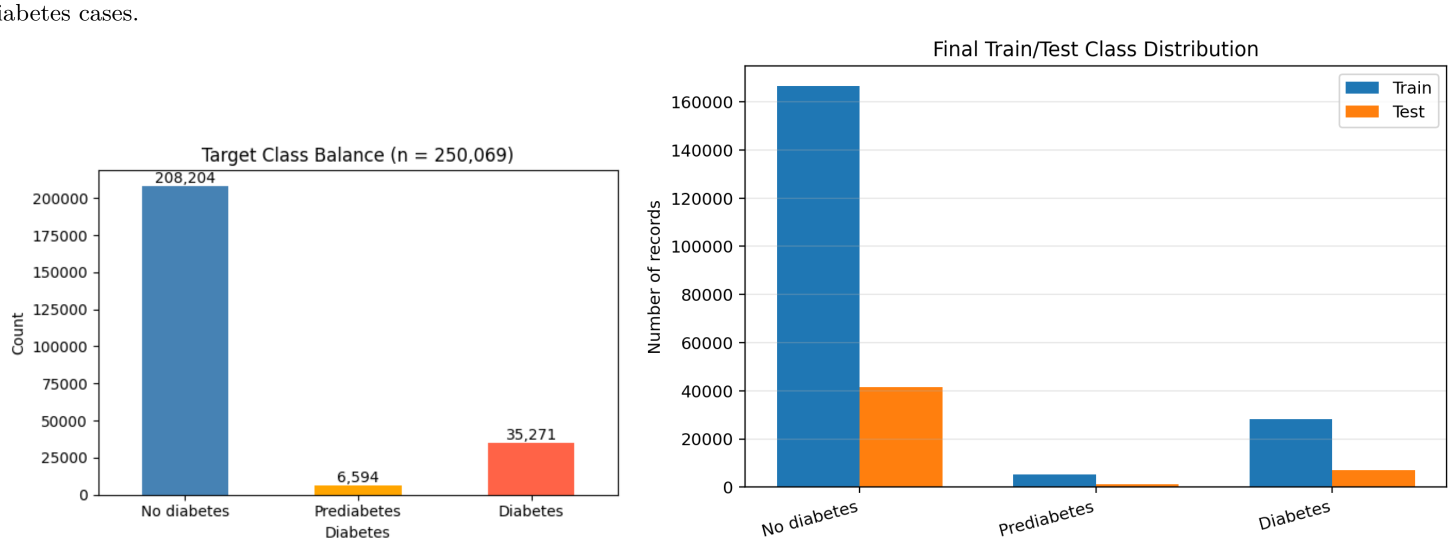
This three-class setup is more realistic than collapsing the task into a binary label because it preserves the intermediate group. However, it makes the problem more difficult because the prediabetes class is small and overlaps clinically with both other classes.

3.3 Cleaning and Recoding

The preprocessing pipeline removed invalid BRFSS codes such as refused, missing, and "do not know" values. Mental-health and physical-health day variables were recoded so that BRFSS code 88 represented zero days. BMI was transformed from the stored BRFSS integer format to a readable continuous value by dividing by 100. Binary variables such as high blood pressure, high cholesterol, stroke, heart disease, physical activity, fruit/vegetable consumption, healthcare access, and difficulty walking were recoded into 0/1 indicators.

3.4 Class Distribution and Imbalance

The final class distribution shows strong imbalance. In the training set, there are 166,563 no-diabetes cases, 5,275 prediabetes cases, and 28,217 diabetes cases. The test set contains 41,641 no-diabetes cases, 1,319 prediabetes cases, and 7,054 diabetes cases. This distribution explains why accuracy alone can be misleading. A model can achieve high accuracy by predicting the majority class but still miss many diabetes cases.



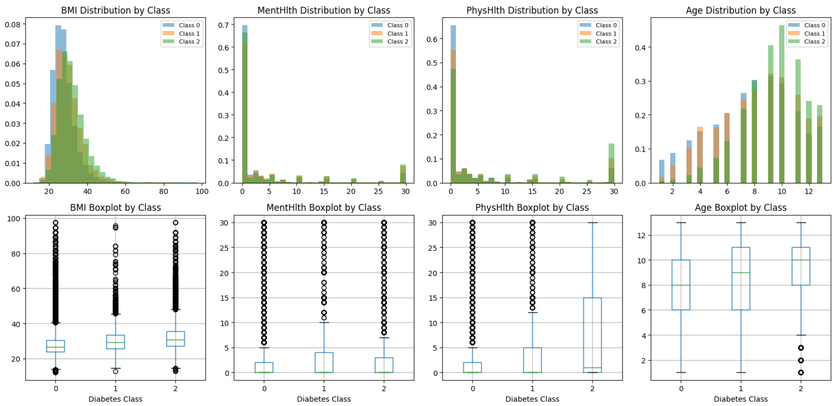
(a) Original EDA target distribution.

(b) Train/test class counts used for modeling.

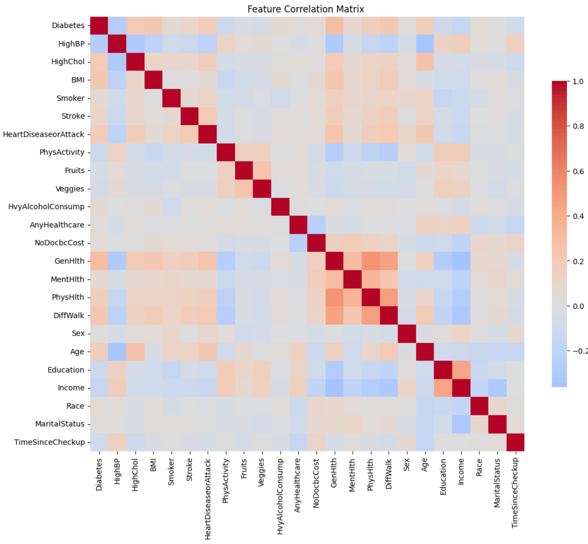
Figure 2: Class imbalance is the central modeling challenge. Prediabetes is the smallest class, and no-diabetes dominates both train and test splits.

3.5 EDA Observations

The EDA showed that BMI, general health, high blood pressure, age, high cholesterol, physical-health days, income, and difficulty walking are strongly related to diabetes status. Continuous variables such as BMI and unhealthy days are skewed, which is common in self-reported health survey data. Correlation analysis showed that several predictors are related to each other, especially general health, physical health, walking difficulty, and cardiovascular variables. This correlation is important when interpreting PDP and SHAP because correlated features can share importance.



(a) Continuous feature distributions.



(b) Correlation heatmap.

Figure 3: EDA visualizations. Diabetes risk indicators show skewed distributions and meaningful correlations, especially among general health, BMI, blood pressure, age, and cardiovascular indicators.

3.6 Feature Engineering

Ten engineered features were created to improve model quality and interpretability:

1. `BMI_class`: obesity-related BMI category.
2. `FINDRISC_score`: diabetes-risk score inspired by age, BMI, physical activity, diet, and high blood pressure.
3. `MetS_score`: metabolic-syndrome-style risk using BMI, blood pressure, and cholesterol.
4. `CV_burden_weighted`: cardiovascular burden from heart disease, stroke, and difficulty walking.
5. `UnhealthyDays_total`: total mental-health and physical-health unhealthy days.
6. `Healthcare_engagement`: healthcare access and checkup behavior.
7. `Protective_lifestyle`: combined physical activity, smoking, diet, and alcohol behavior score.
8. `SES_risk_score`: socioeconomic and healthcare-access risk.
9. `BMI_x_Age`: interaction between BMI and age group.
10. `GenHlth_x_DiffWalk`: interaction between general health and walking difficulty.

3.7 Feature Selection

Feature selection combined variance filtering, VIF awareness, ANOVA F-score, mutual information, chi-square scores, random-forest importance, permutation importance, and logistic-regression coefficient magnitude. The final 20 predictors were `GenHlth`, `HighBP`, `GenHlth_x_DiffWalk`, `BMI_x_Age`, `CV_burden_weighted`, `HighChol`, `DiffWalk`, `Sex`, `BMI_class`, `SES_risk_score`, `Income`, `PhysActivity`, `HeartDiseaseorAttack`, `HvyAlcoholConsump`, `MetS_score`, `BMI`, `FINDRISC_score`, `PhysHlth`, `TimeSinceCheckup`, and `Age`.

4 Methodology

4.1 Implemented Models

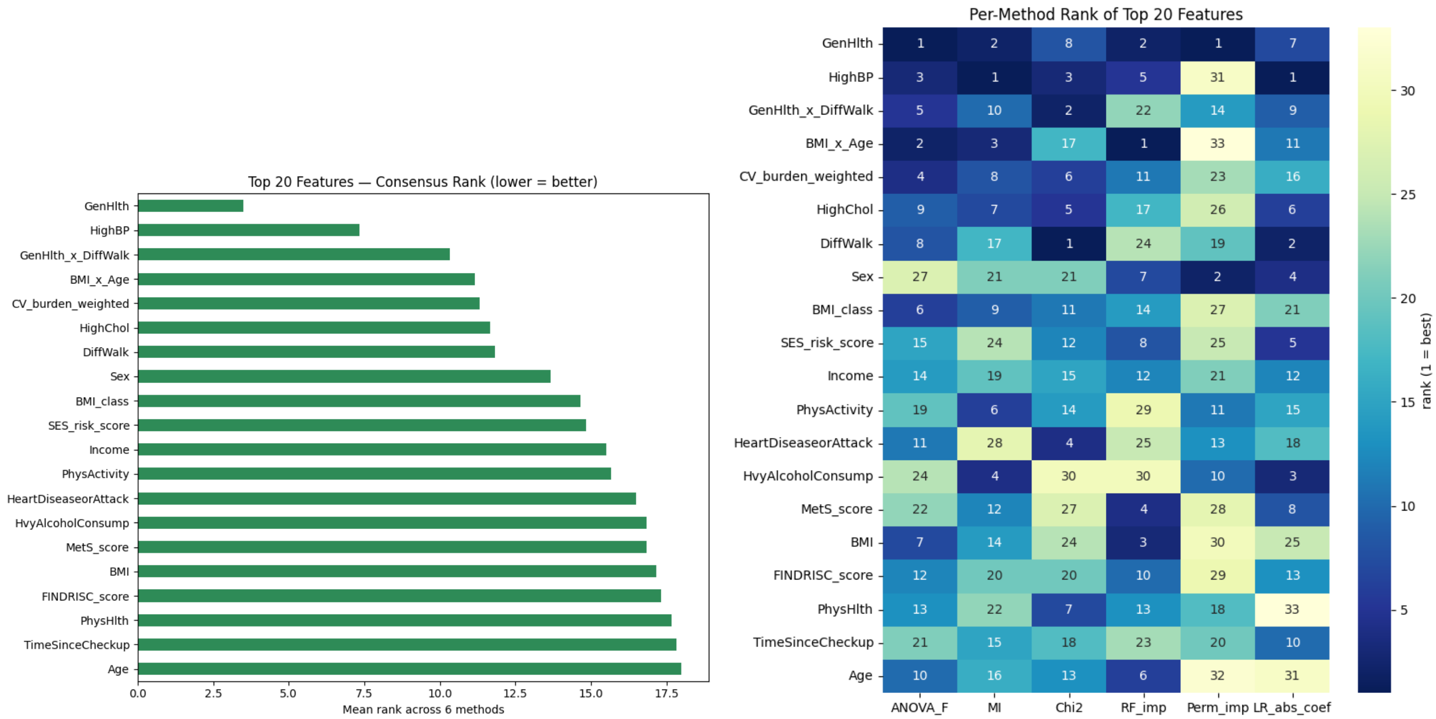
Six final model tracks were implemented. The models were selected from the best or most representative model direction of each assigned paper, then adapted to the final BRFSS multiclass feature table. Table **??** summarizes the tracks.

Table 2: Final implemented models and implementation decisions.

Member	**Paper Track**	**Implemented Model**	**Reason for selection and adaptation**
Dawood	Xie et al.	MLP / neural network	Xie reported the neural network as the highest-accuracy/AUC model. Our MLP keeps this high-accuracy direction.
Dawood	Ullah et al.	XGBoost	Ullah emphasized imbalance and high-risk-factor detection. XGBoost was used as a stronger, more stable ensemble alternative for the final three-class task.
Dawood	Chowdhury et al.	Histogram Gradient Boosting	Chowdhury focused on imbalanced BRFSS learning. HistGB preserves the boosting/imbalance direction without destructive resampling.
Youssef	Islam et al.	Extra Trees	Islam reported Extra Trees as a strong BRFSS multiclass model with SHAP/PDP/LIME explainability.
Youssef	Zafar et al.	TIPNet-inspired deep tabular model	Zafar proposed TIPNet. The implementation adapts the idea to tabular survey features because the available data are not longitudinal.
Youssef	Maimaitijiang et al.	CatBoost	Maimaitijiang used CatBoost and SHAP for explainable online diabetes-risk prediction.

4.2 Imbalance-Handling Strategy

Early experiments showed that aggressive resampling can damage test accuracy. Methods such as SMOTE-ENN and ENN may improve minority-class recall but can also distort the training distribution and produce excessive false positives when evaluated on the naturally imbalanced BRFSS test set. Therefore, the final models use a more conservative strategy: high-accuracy default predictions, class weights or soft sample weights, balanced subsampling where appropriate, and threshold analysis for screening-oriented interpretation. This allows the



(a) Feature-selection ranking. (b) Selected-feature importance view.

Figure 4: Feature-selection outputs. The final feature set balances statistical relevance, model utility, and clinical interpretability.

paper to report both realistic accuracy and diabetes recall instead of optimizing only one metric.

4.3 Hyperparameter and Optimization Decisions

The final models use practical hyperparameters chosen to balance runtime, stability, and performance. XGBoost and CatBoost use moderate tree depth, learning rates below 0.05, hundreds of boosting iterations, subsampling, and regularization to reduce overfitting. Extra Trees uses a large number of trees, limited minimum leaf size, and balanced subsampling. The deep tabular model uses batch normalization, dropout, early stopping, and class weights. These choices make the notebooks runnable while keeping the model behavior appropriate for a large imbalanced tabular dataset.

4.4 Evaluation Metrics

The main evaluation metrics are accuracy, balanced accuracy, macro-F1, weighted-F1, ROC-AUC, and diabetes recall. Accuracy measures overall correctness under the natural test distribution. Balanced accuracy and macro-F1 reduce the dominance of the majority class. Weighted-F1 summarizes performance while respecting class frequencies. Diabetes recall measures the ability to detect true diabetes cases, which is important for screening. ROC-AUC evaluates ranking quality for multiclass one-vs-rest probability outputs.

4.5 Explainability Techniques

A unified explanation framework is used for all models. This allows fair comparison of model behavior. The six XAI tools are:

- **Permutation Feature Importance (PFI):** global importance measured by performance decrease after shuffling a feature.
- **Partial Dependence Plot (PDP):** average effect of changing a feature while keeping other features fixed.
- **Individual Conditional Expectation (ICE):** individual feature-response curves that show heterogeneity hidden by PDP averages.
- **SHAP:** additive feature attribution for global and local explanation.
- **LIME:** local surrogate explanation for a selected individual prediction.
- **Global Surrogate Tree:** a shallow interpretable tree trained to approximate the predictions of a black-box model.

Table 3: Comparison of XAI techniques used in the final implementation.

XAI tool	Scope	Main output	Strengths and limitations
PFI	Global	Ranked features by performance impact	Easy to compare across models; may undervalue correlated features.
PDP	Global	Average feature-response curve	Shows broad trends; can be misleading when features are strongly correlated.
ICE	Local/global hybrid	Individual response curves	Reveals heterogeneity; can be visually dense with many samples.
SHAP	Global and local	Additive contribution values	Strong attribution framework; computationally expensive for model-agnostic use.
LIME	Local	Local linear/surrogate explanation	Easy to explain for one case; can vary depending on perturbation settings.
Surrogate tree	Global approximation	Simple decision tree rules	Very readable; only valid when fidelity to the original model is acceptable.

5 Results

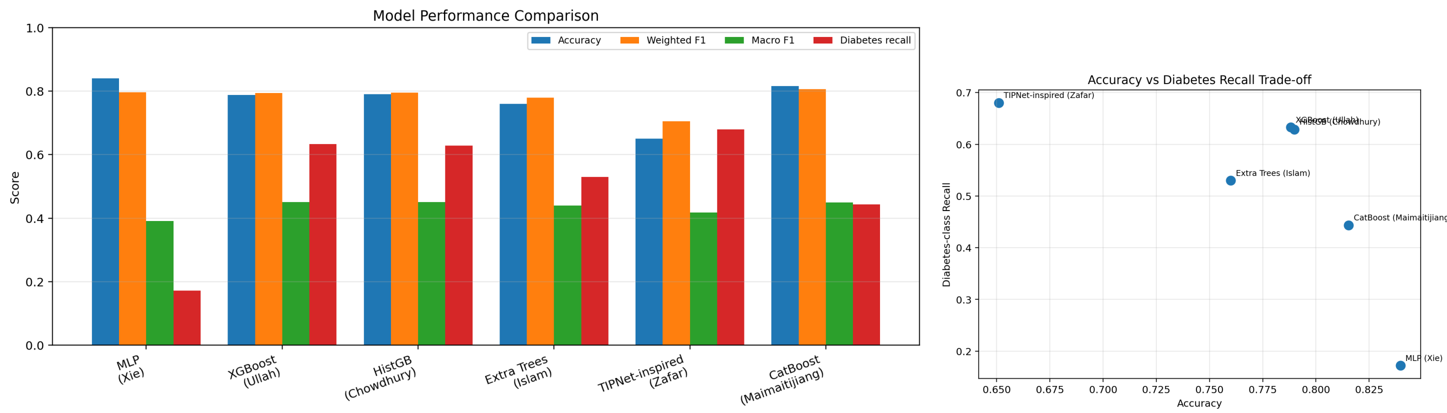
5.1 Overall Performance Comparison

Table ?? reports the final test-set outputs from the executed notebooks. The results show that there is no single best model for all goals. The MLP has the strongest accuracy, but low diabetes recall. XGBoost and HistGB improve diabetes recall substantially, but their accuracy is lower. CatBoost provides strong weighted-F1 and stable overall performance. The TIPNet-inspired model is the most recall-oriented but sacrifices accuracy.

Table 4: Final test-set results from the executed notebooks.

Member	Paper	Model	Acc.	Bal. Acc.	Macro-F1	W-F1	AUC-W	Diab. Rec.
Dawood	Xie	MLP	0.8397	0.3839	0.3917	0.7960	0.8083	0.1722
Dawood	Ullah	XGBoost	0.7881	0.4910	0.4511	0.7945	0.8102	0.6331
Dawood	Chowdhury	HistGB	0.7899	0.4904	0.4514	0.7957	0.8092	0.6289
Youssef	Islam	Extra Trees	0.7600	0.4600	0.4400	0.7800	0.7051	0.5300
Youssef	Zafar	TIPNet-inspired	0.6510	0.4900	0.4183	0.7050	–	0.6800
Youssef	Maimaitijiang	CatBoost	0.8154	0.4527	0.4497	0.8065	0.7781	0.4437

W-F1 = weighted F1-score; AUC-W = weighted ROC-AUC; Diab. Rec. = diabetes-class recall.



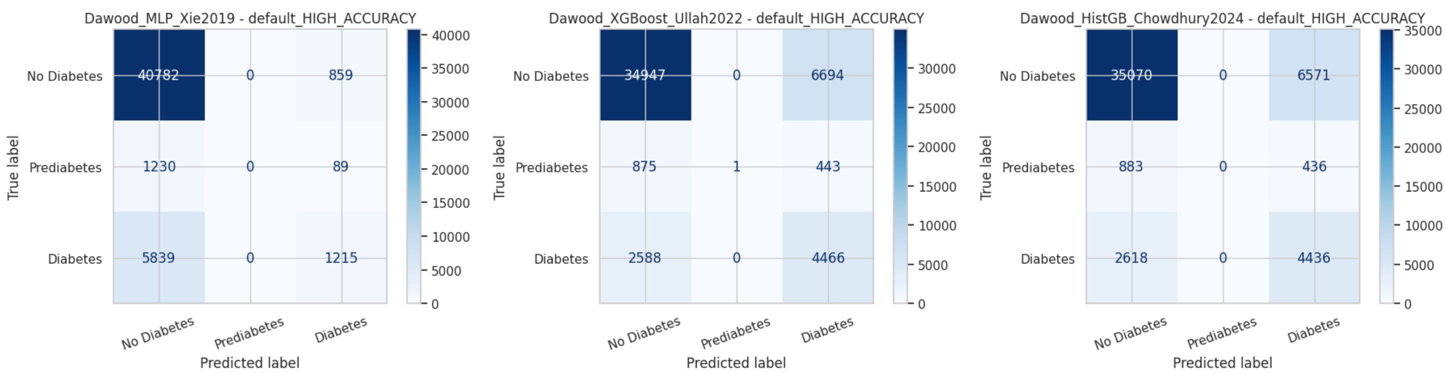
(a) Grouped performance comparison.

(b) Accuracy-recall trade-off.

Figure 5: Model comparison visuals. High accuracy and high diabetes recall are not always achieved by the same model. This is the central trade-off in the final results.

5.2 Dawood Model Results

The Xie-track MLP achieved the highest overall accuracy among all models. However, its diabetes recall is low, meaning it is strong at recognizing the majority no-diabetes class but weak as a screening tool. The Ullah-track XGBoost and Chowdhury-track HistGB models achieved lower accuracy but much higher diabetes recall. Their behavior is more useful when the goal is to detect diabetes cases for follow-up screening. This confirms that the choice of best model depends on the deployment objective.



(a) MLP confusion matrix.

(b) XGBoost confusion matrix.

(c) HistGB confusion matrix.

Figure 6: Dawood model evaluation outputs. The MLP favors majority-class accuracy, while XGBoost and HistGB improve diabetes-case detection.

5.3 Youssef Model Results

The Islam-track Extra Trees model follows a paper with strong BRFSS multiclass results and provides an interpretable tree-ensemble baseline. The Zafar-track TIPNet-inspired model has lower accuracy but higher diabetes recall, showing that class-aware deep training can increase sensitivity at the cost of false positives. The Maimaitijiang-track CatBoost model achieves the strongest weighted-F1 among Youssef's models and a high accuracy of 0.8154, making it the best overall Youssef track under the natural test distribution.

5.4 Model Comparison Interpretation

The model comparison reveals a repeated pattern: high-accuracy models tend to be conservative, while recall-oriented models predict diabetes more often. In a real screening scenario, the cost of false negatives may be higher than the cost of false positives because a missed diabetic patient may not receive follow-up testing. However, too many false positives may waste clinical resources and create patient anxiety. Therefore, a practical system would not select a model based only on accuracy; it would choose a threshold and model family based on the intended use case.

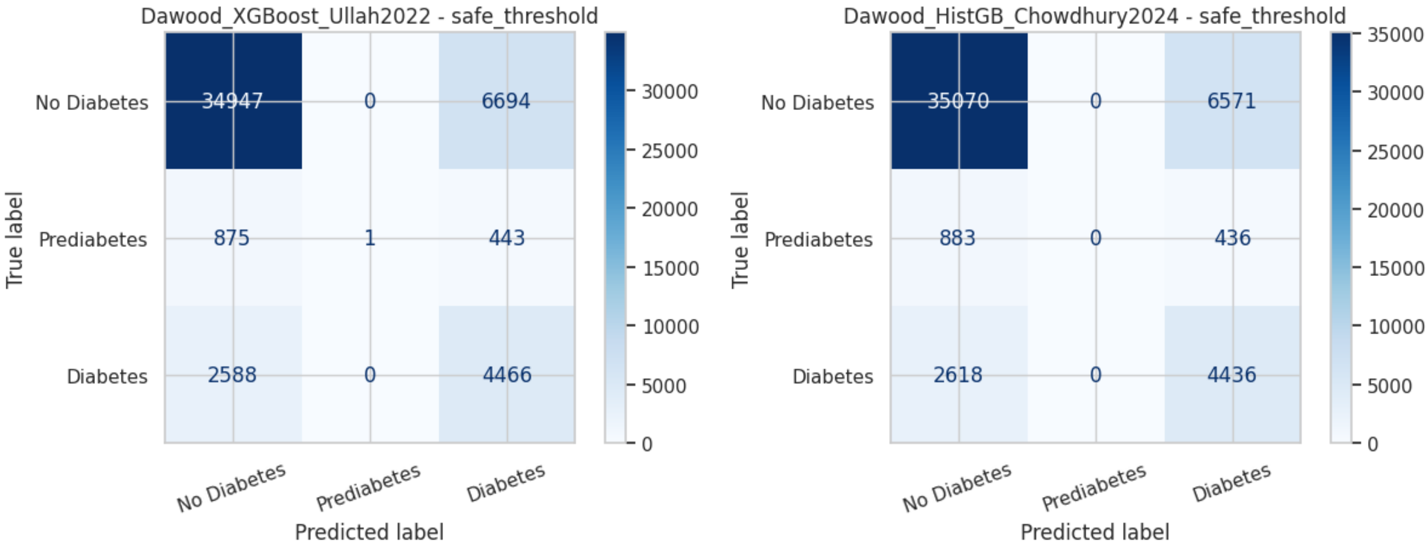
6 Explainability Results

6.1 Global Feature Importance

PFI and SHAP outputs repeatedly identify general health, high blood pressure, BMI-related variables, age, high cholesterol, checkup frequency, and metabolic-risk indicators as important. This consistency across model families increases confidence that the models are learning medically meaningful patterns rather than random artifacts.

6.2 PDP and ICE

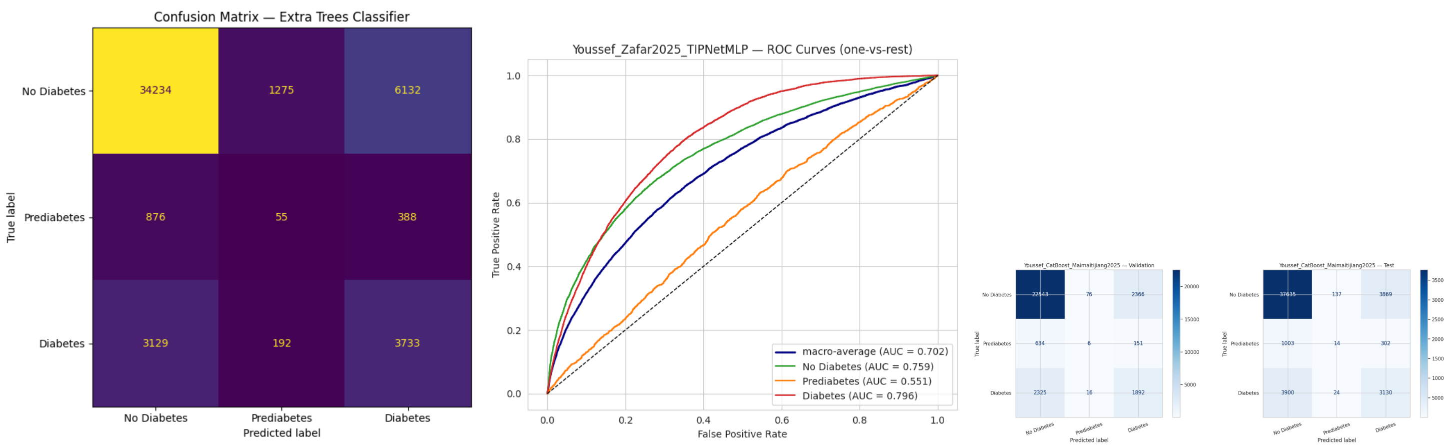
PDP and ICE answer different questions. PDP shows the average effect of a feature on predicted diabetes probability. ICE shows how that effect changes across individual records. In this project, PDP/ICE are especially useful for BMI, age, general health, and high blood pressure.



(a) XGBoost ROC/PR-style output.

(b) HistGB ROC/PR-style output.

Figure 7: Additional Dawood evaluation visuals. These outputs support the interpretation that tree-boosting models provide better probability ranking for diabetes-related screening than accuracy alone suggests.

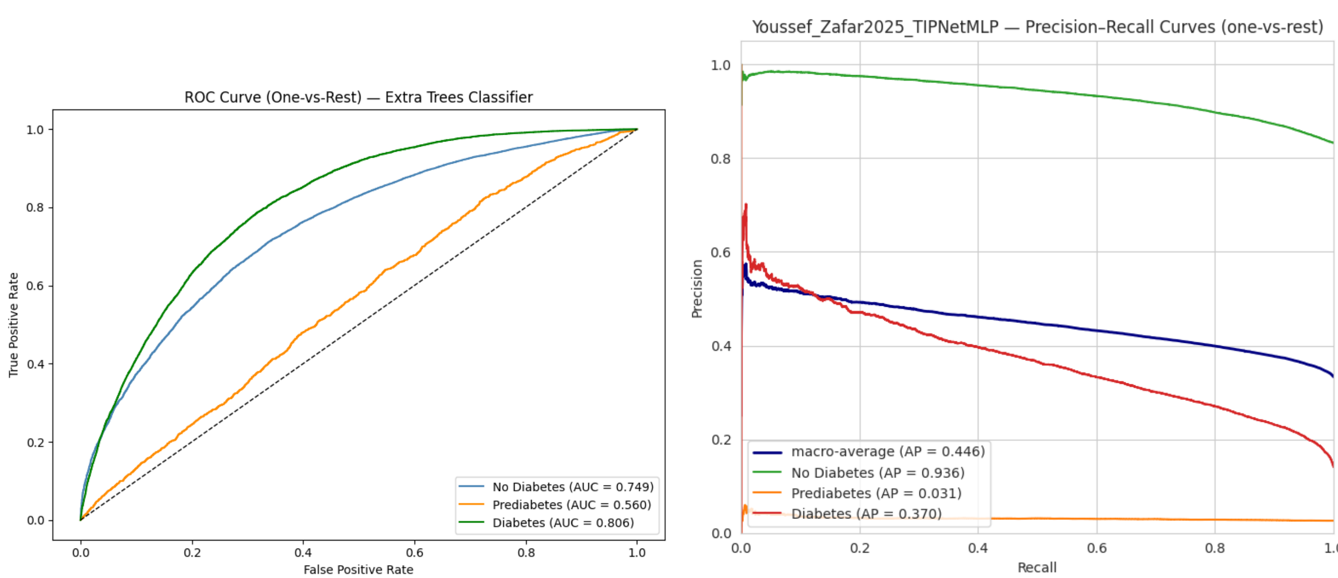


(a) Extra Trees confusion matrix.

(b) TIPNet-inspired confusion matrix.

(c) CatBoost evaluation output.

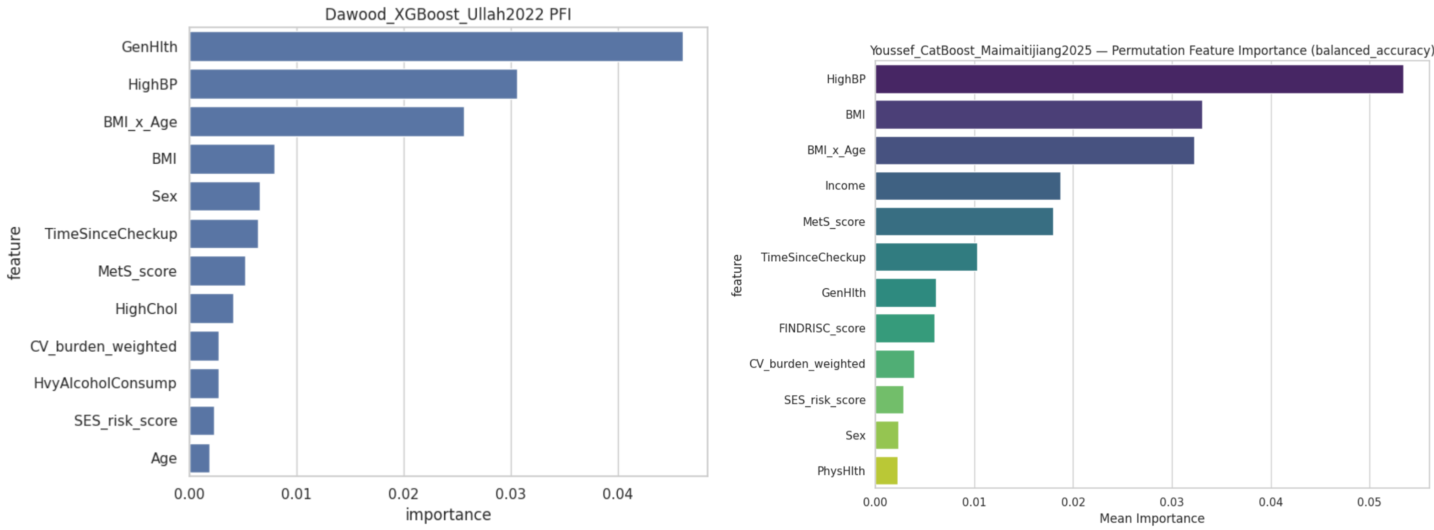
Figure 8: Youssef model evaluation outputs. CatBoost is the strongest overall model, while TIPNet-inspired prediction is more screening-sensitive.



(a) Extra Trees ROC curves.

(b) TIPNet-inspired ROC curves.

Figure 9: Youssef probability-ranking outputs. ROC curves help evaluate models beyond simple class labels.

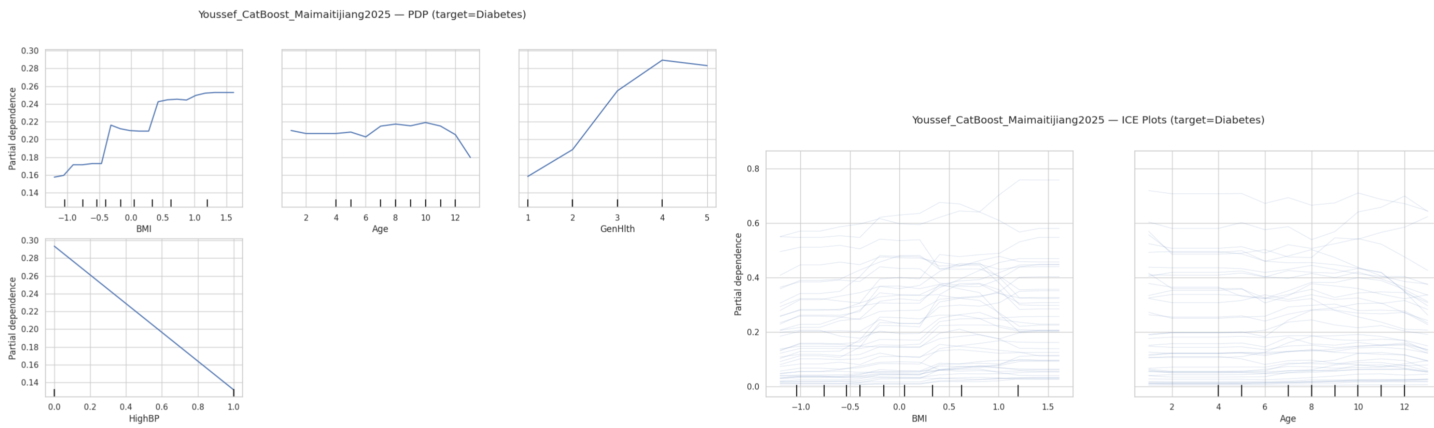


(a) XGBoost permutation importance.

(b) CatBoost permutation importance.

Figure 10: Global feature-importance examples. General health, high blood pressure, BMI-related indicators, and checkup-related variables appear repeatedly.

For example, average risk generally rises with worse general health and higher metabolic burden, but ICE curves show that individuals with different cardiovascular and socioeconomic profiles can respond differently.



(a) CatBoost PDP.

(b) CatBoost ICE.

Figure 11: PDP and ICE comparison. PDP summarizes average model behavior, while ICE reveals case-level heterogeneity.

6.3 SHAP and LIME

SHAP provides additive feature attributions and is suitable for both global summaries and local prediction explanation. LIME creates a local surrogate model for a specific prediction. In the final notebooks, SHAP is more useful for comparing feature contribution patterns across records, while LIME is more useful for explaining one selected case in a human-readable way. The two tools are complementary rather than interchangeable.

6.4 Surrogate Trees

Global surrogate trees approximate a complex model with a shallow decision tree. The purpose is not to replace the original model but to summarize its main decision structure. Surrogate trees are useful for presentation and stakeholder communication, but the fidelity score must be checked. If fidelity is low, the surrogate should be treated only as an approximate explanation.

7 Discussion

7.1 Key Findings

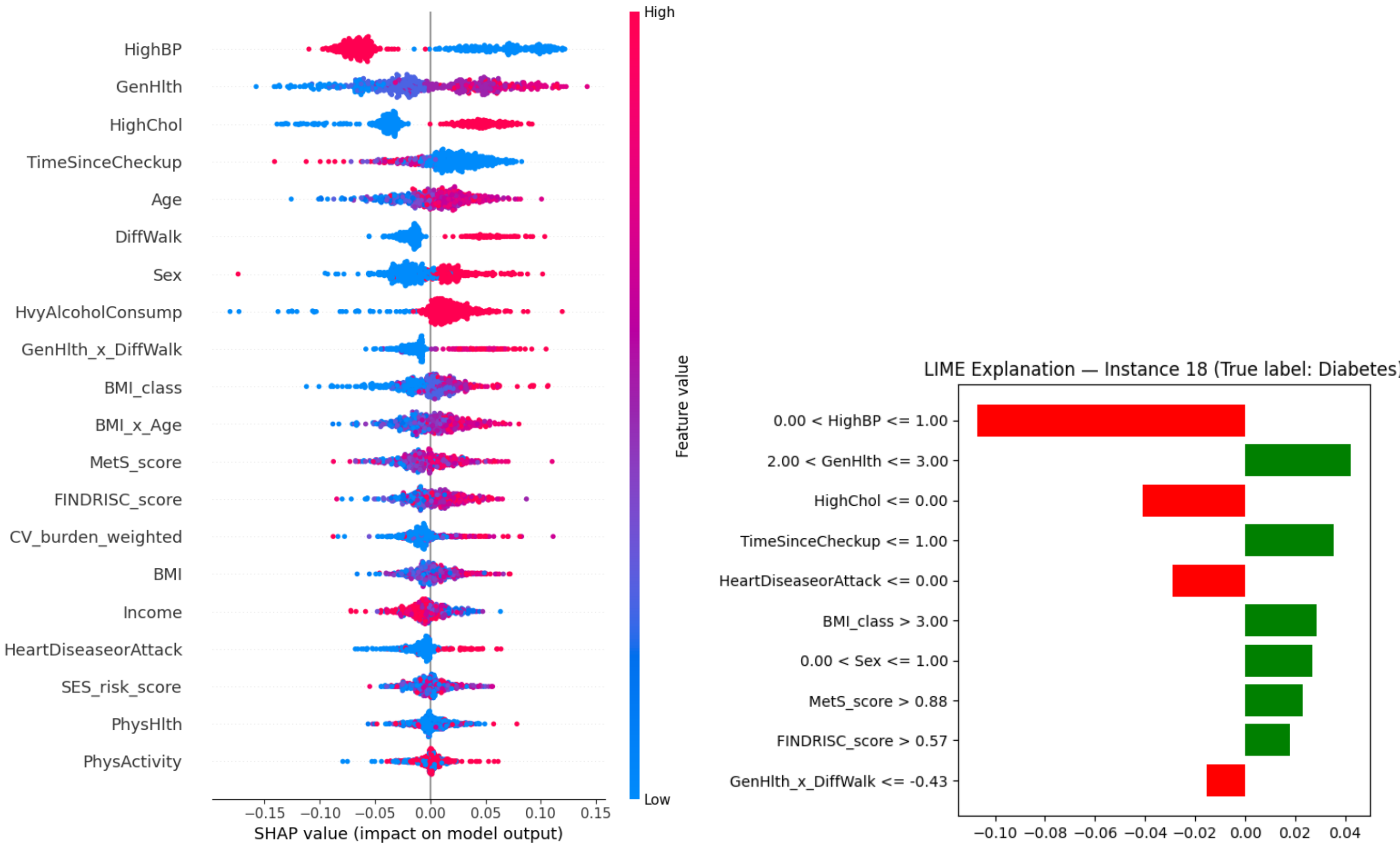
The first major finding is that performance depends strongly on the chosen evaluation goal. If the goal is highest overall accuracy, the MLP and CatBoost are preferred. If the goal is stronger diabetes detection, XGBoost, HistGB, and the TIPNet-inspired model are more useful. This is expected in an imbalanced medical screening problem because no single metric captures all clinical needs.

The second finding is that prediabetes is the hardest class. Prediabetes is small in the dataset and clinically intermediate: it shares characteristics with both no-diabetes and diabetes. This causes models to confuse prediabetes with the other classes. A future project could treat the problem as ordinal classification or use a two-stage pipeline: first detect any diabetes-related risk, then distinguish prediabetes from diabetes.

The third finding is that XAI outputs are reasonably stable across models. General health, high blood pressure, BMI-related variables, age, high cholesterol, and checkup behavior appear repeatedly. This supports the credibility of the models because these features are consistent with known diabetes risk factors.

7.2 Why Published Results May Be Higher

Some papers report much higher accuracy than the results in this project. This difference does not necessarily mean our implementation is wrong. Several methodological differences explain the gap. First, some papers use binary targets, while this project keeps a harder three-class target. Second, some results are reported on balanced or resampled evaluation sets, while this project evaluates on a naturally imbalanced held-out test set. Third, some papers use different feature sets, years, or cleaned versions of BRFSS. Fourth, deep-learning papers may use cross-validation and more extensive hyperparameter optimization. Our results are conservative because they preserve the natural class imbalance and report diabetes recall explicitly.



(a) Extra Trees SHAP diabetes-class explanation.

(b) Extra Trees LIME local explanation.

Figure 12: SHAP and LIME examples. SHAP provides structured attribution values, while LIME provides a local case-level explanation.

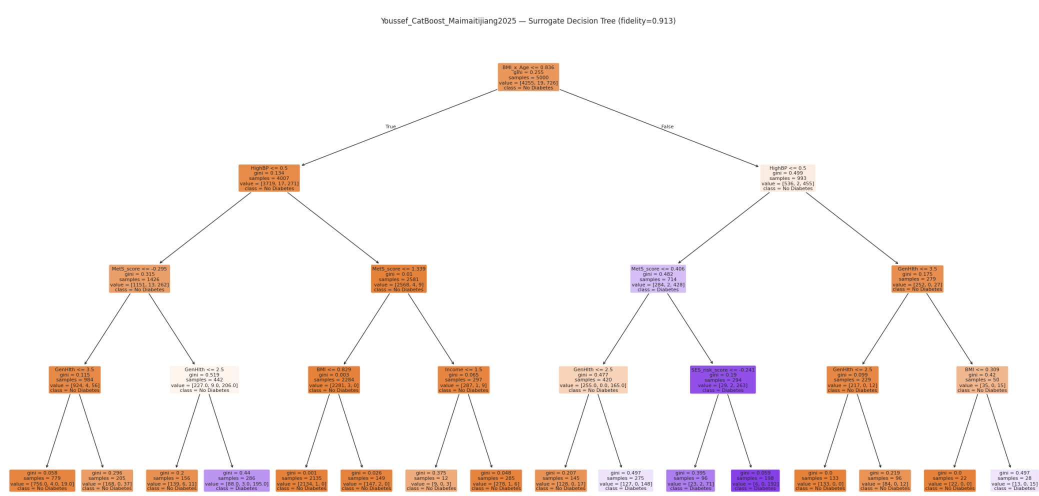


Figure 13: Example global surrogate tree for CatBoost. The tree provides a simplified approximation of the black-box model and should be interpreted together with its fidelity value.

7.3 Imbalance and Threshold Trade-off

Class imbalance is the main technical challenge. Aggressive resampling methods can improve minority recall but may reduce accuracy by increasing false positives. Conservative class weights keep accuracy stable but may miss minority cases. The correct strategy depends on use case. For public-health screening, a lower diabetes threshold may be acceptable if it leads to follow-up testing rather than final diagnosis. For automatic classification, a conservative threshold is safer. Therefore, the final report presents both accuracy and diabetes recall instead of hiding this trade-off.

7.4 Practical Implications

The project is best understood as a screening-support system, not a diagnostic system. A model could help prioritize individuals for follow-up testing or health education, especially when laboratory testing is not immediately available. The explanations make the outputs more useful: instead of saying only “high risk”, the model can show that the risk is driven by poor general health, high blood pressure, BMI, or checkup history. This supports more transparent communication and helps avoid treating the model as an unexplained black box.

7.5 Limitations

The dataset is self-reported, so it may contain recall bias and reporting bias. The target is also self-reported rather than laboratory-confirmed. The feature set does not include glucose, HbA1c, insulin, or medication history. Prediabetes is very underrepresented. Some paper reproductions are adaptations rather than exact reproductions because the available BRFSS feature table differs from the data used in each paper. Finally, the models were not externally validated on another BRFSS year or a clinical EHR dataset.

8 Conclusion

This study developed a complete explainable diabetes prediction framework using BRFSS 2015 health indicators. The pipeline includes raw-data preprocessing, EDA, feature engineering, feature selection, six paper-inspired model tracks, and a unified XAI framework. The MLP achieved the strongest overall accuracy, XGBoost and HistGB provided stronger diabetes recall, CatBoost provided strong weighted performance, and the TIPNet-inspired model showed high screening sensitivity but lower accuracy. Across XAI tools, general health, high blood pressure, BMI, age, high cholesterol, and checkup-related features repeatedly appeared as important predictors. The main conclusion is that explainability is necessary for responsible healthcare AI because it helps interpret model behavior, compare models beyond accuracy, and communicate risk factors transparently. Future work should test external BRFSS years, improve prediabetes detection, evaluate fairness across demographic groups, compare binary and ordinal formulations, and design clinically justified threshold policies for screening.

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